ROYAL UNIVERSITY OF BHUTAN

COLLEGE OF SCIENCE AND TECHNOLOGY

NWC201_Computer Communication Network

Module Tutor: Nima

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Department: 1st year Software Engineering

Date of Submission: 22nd October, 2025

Due: 22nd October, 2025

1. AIM

To learn how to configure and test standard and extended Access Control Lists (ACLs) on a Cisco router.

2. OBJECTIVES

- Understand what ACLs are and how they work.
- Configure standard and extended ACLs using Cisco Packet tracer
- Apply ACLs to router interfaces
- Test and verify the results

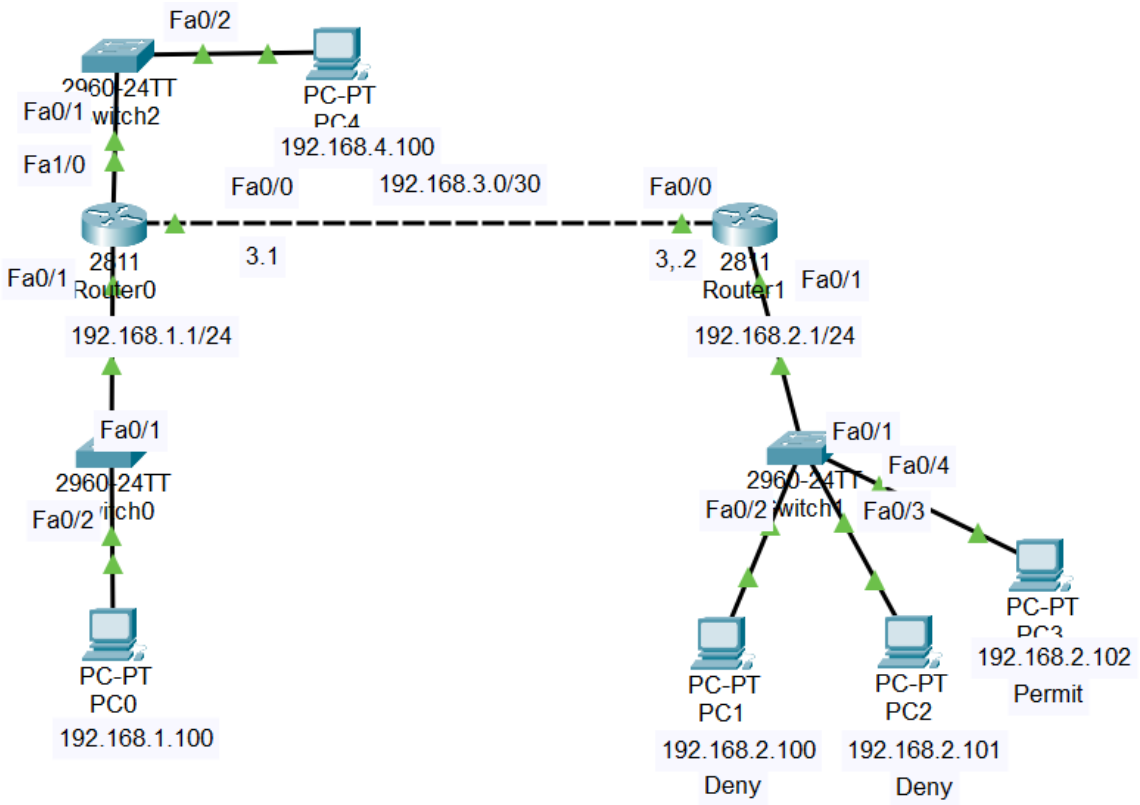
3. BACKGROUND

Access Control Lists (ACLs) in Cisco routers help manage network traffic by permitting or blocking data packets according to specific criteria like IP addresses, protocols, or port numbers.

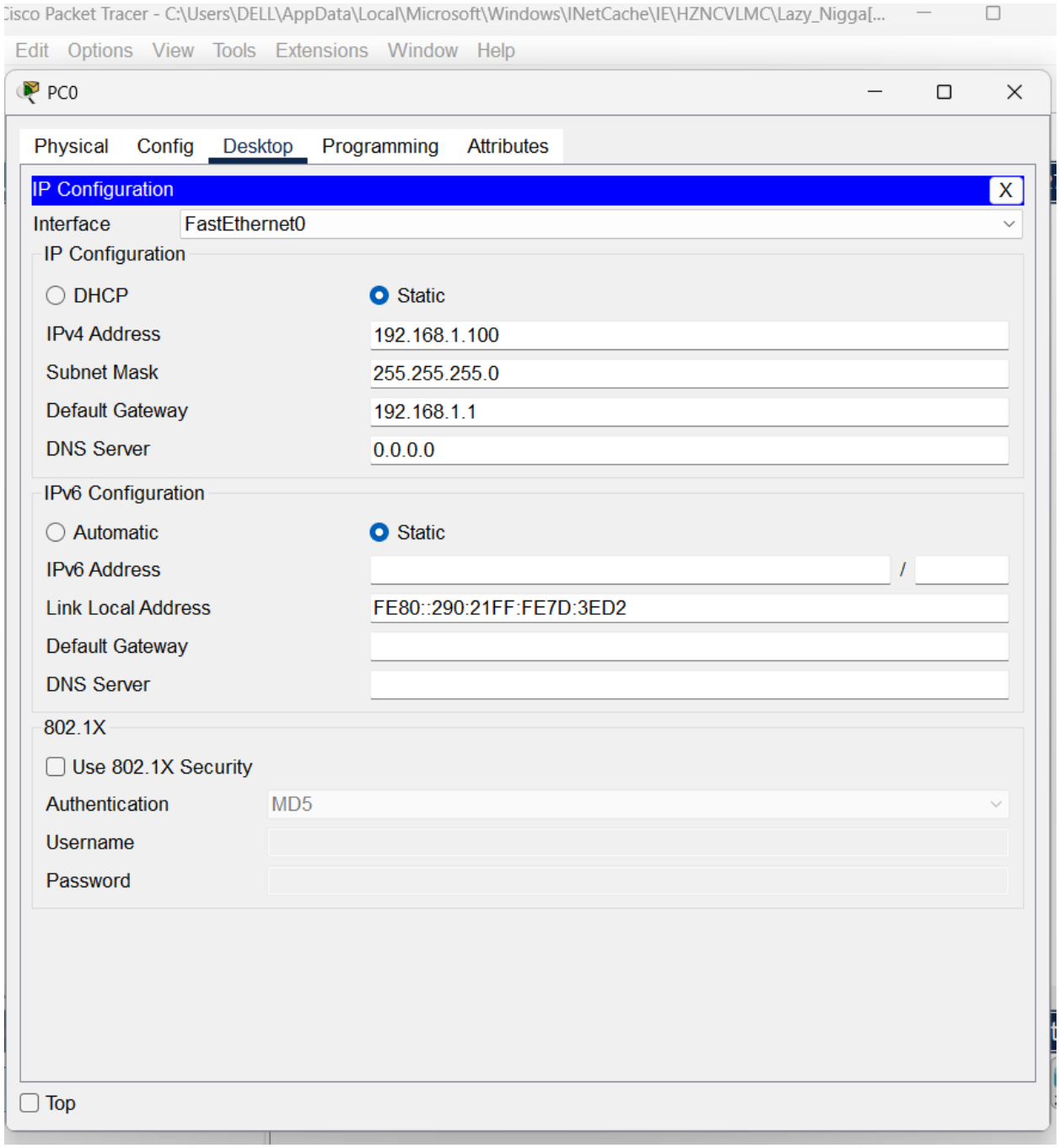
- **Standard ACLs** restrict traffic solely based on the source IP address.
- **Extended ACLs** provide more granular control by filtering traffic based on source and destination IP addresses, as well as protocol types and port numbers.

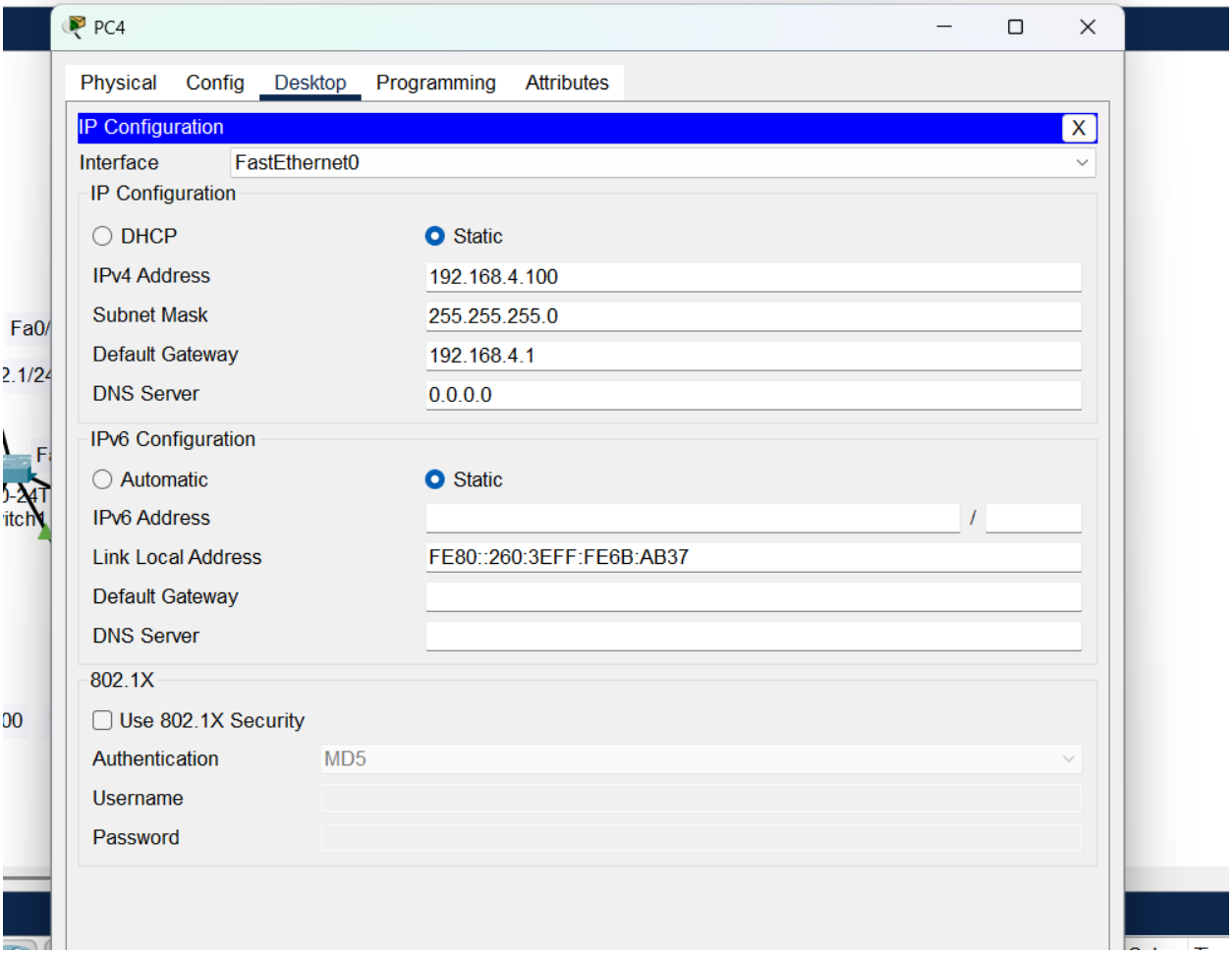
4. PROCEDURES TASK

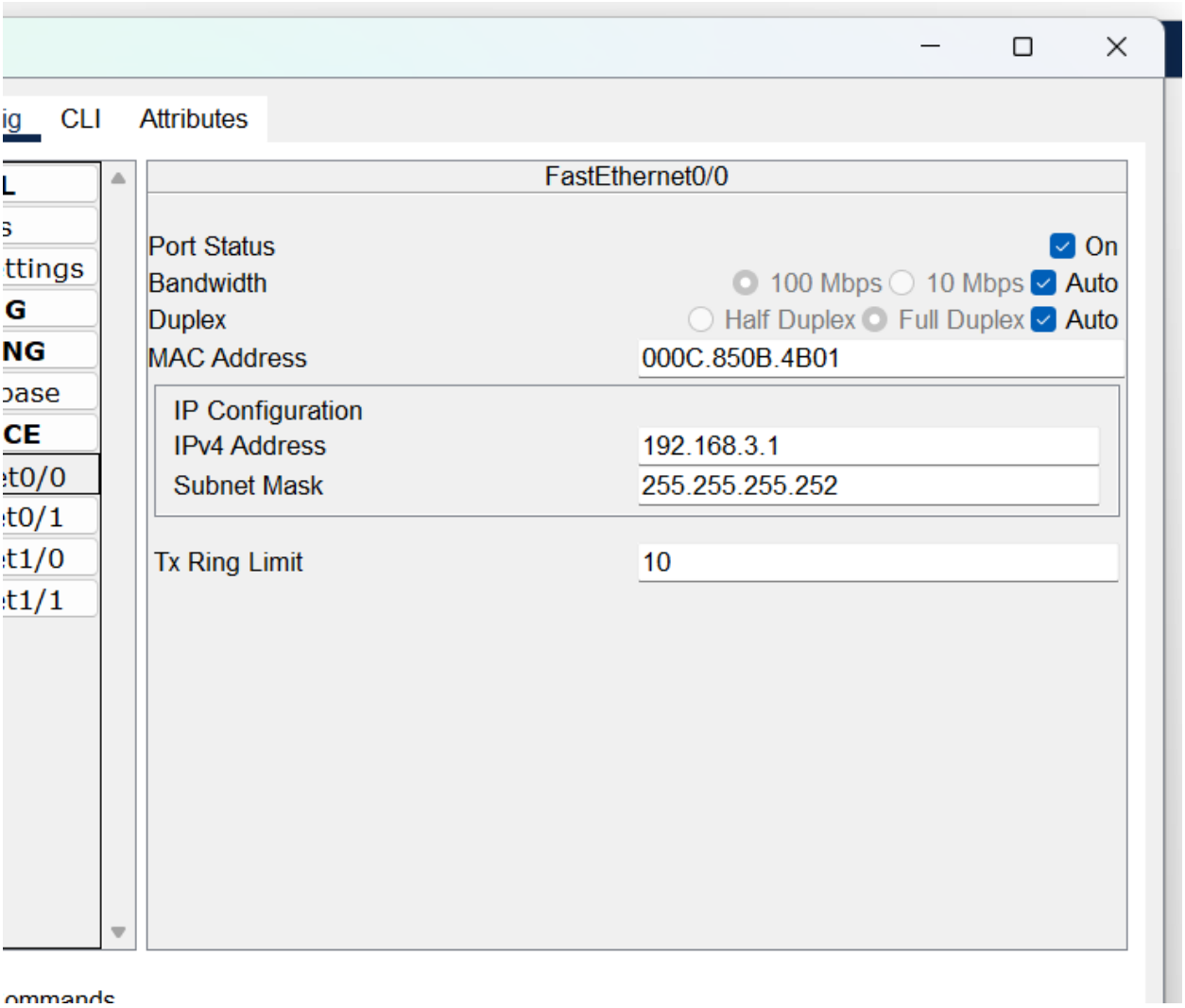
1. Create a simple network with two router, three switch and five PC



3. Assign IP addresses to all devices.




PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.4.100
Subnet Mask
255.255.255.0
Default Gateway
192.168.4.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:3EFF:FE6B:AB37
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password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CLI
Attributes
L
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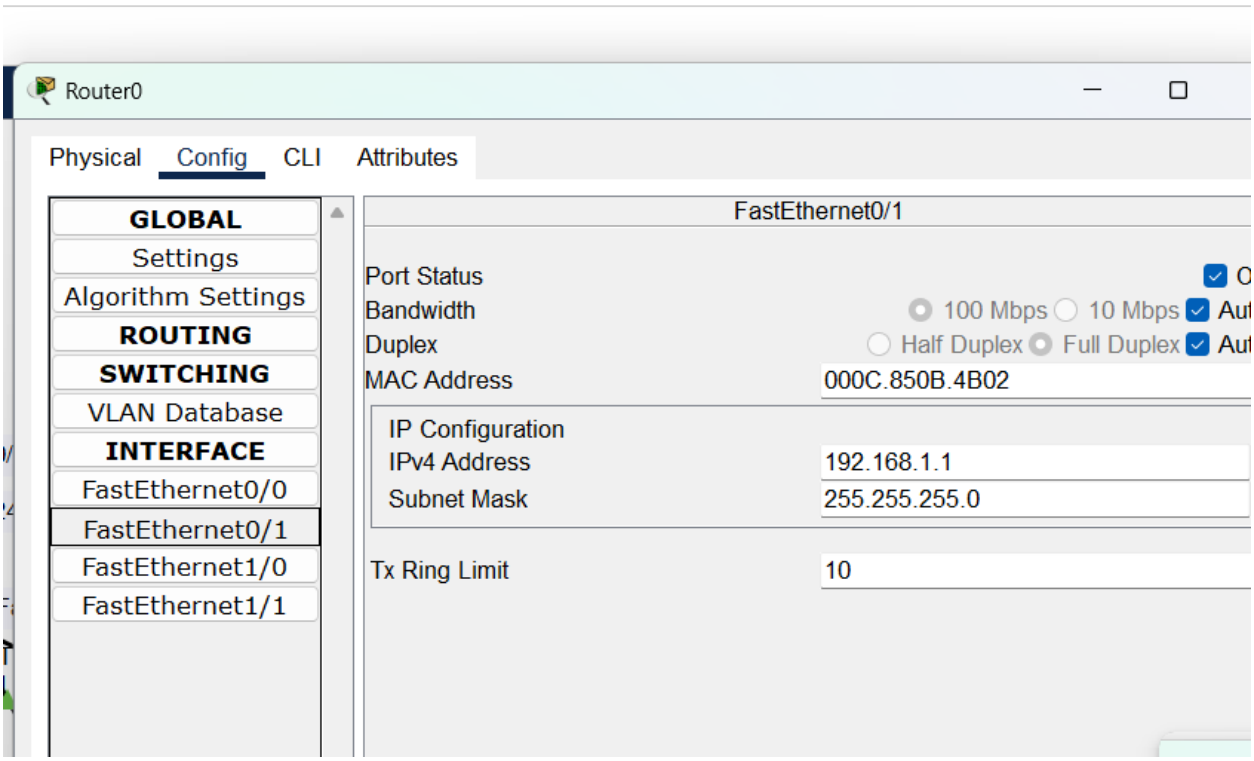
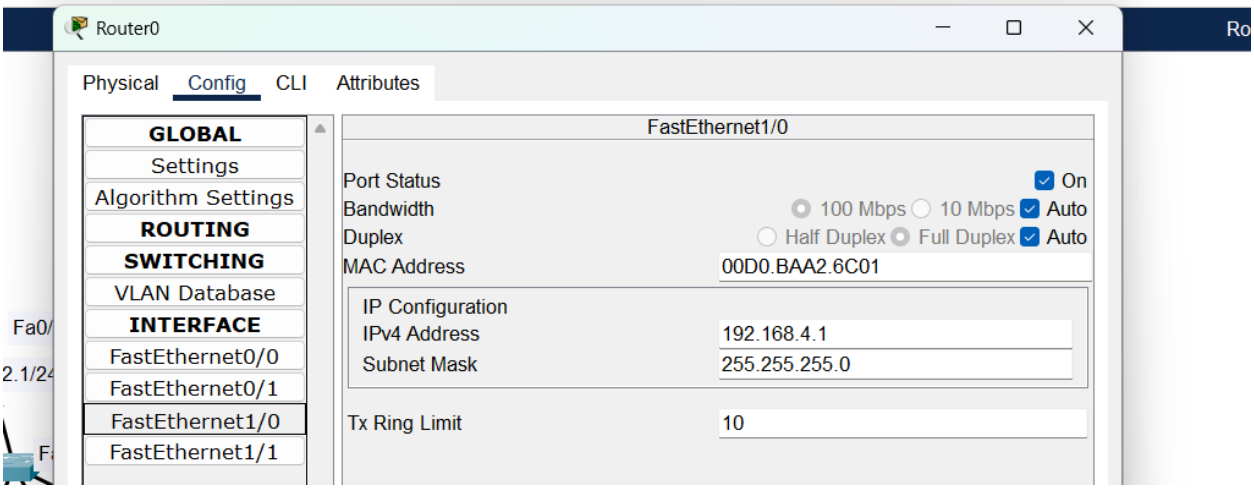
ttings
G
NG
base
CE
et0/0
et0/1
et1/0
et1/1
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000C.850B.4B01
IP Configuration
IPv4 Address
192.168.3.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10
ommands





3. Configure a Standard ACL:

```
access-list 1 deny 192.168.2.100
access-list 1 permit any
interface fa0/0
ip access-group 99 out
```

Physical Config CLI Attributes

IOS Command Line Interface

```
Model number                     : WS-C2960-24TT-L
System serial number             : FOC1010X104
Top Assembly Part Number         : 800-27221-02
Top Assembly Revision Number     : A0
Version ID                       : V02
CLEI Code Number                 : COM3L00BRA
Hardware Board Revision Number   : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#
```

Copy Paste

PC4

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.4.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.4.1

DNS Server 0.0.0.0

IPv6 Configuration

Automatic Static

IPv6 Address /

Link Local Address FE80::260:3EFF:FE6B:AB37

Default Gateway

DNS Server

802.1X

IOS Command Line Interface

```
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 ip access-group 99 out
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.2.0 255.255.255.0 192.168.3.2
!
ip flow-export version 9
!
!
access-list 99 deny host 192.168.2.100
access-list 99 deny host 192.168.2.101
access-list 99 permit any
```

Co

4. Configure an Extended ACL:
   access-list 101 permit ip access-group 99 out

5. Test with ping and telnet commands to see which traffic is allowed or denied.

PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.100

Pinging 192.168.2.100 with 32 bytes of data:

Reply from 192.168.2.100: bytes=32 time<1ms TTL=128
Reply from 192.168.2.100: bytes=32 time<1ms TTL=128
Reply from 192.168.2.100: bytes=32 time<1ms TTL=128
Reply from 192.168.2.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC3

Physical | Config | Desktop | Programming | Attributes

Command Prompt

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Ping statistics for 192.168.2.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.101

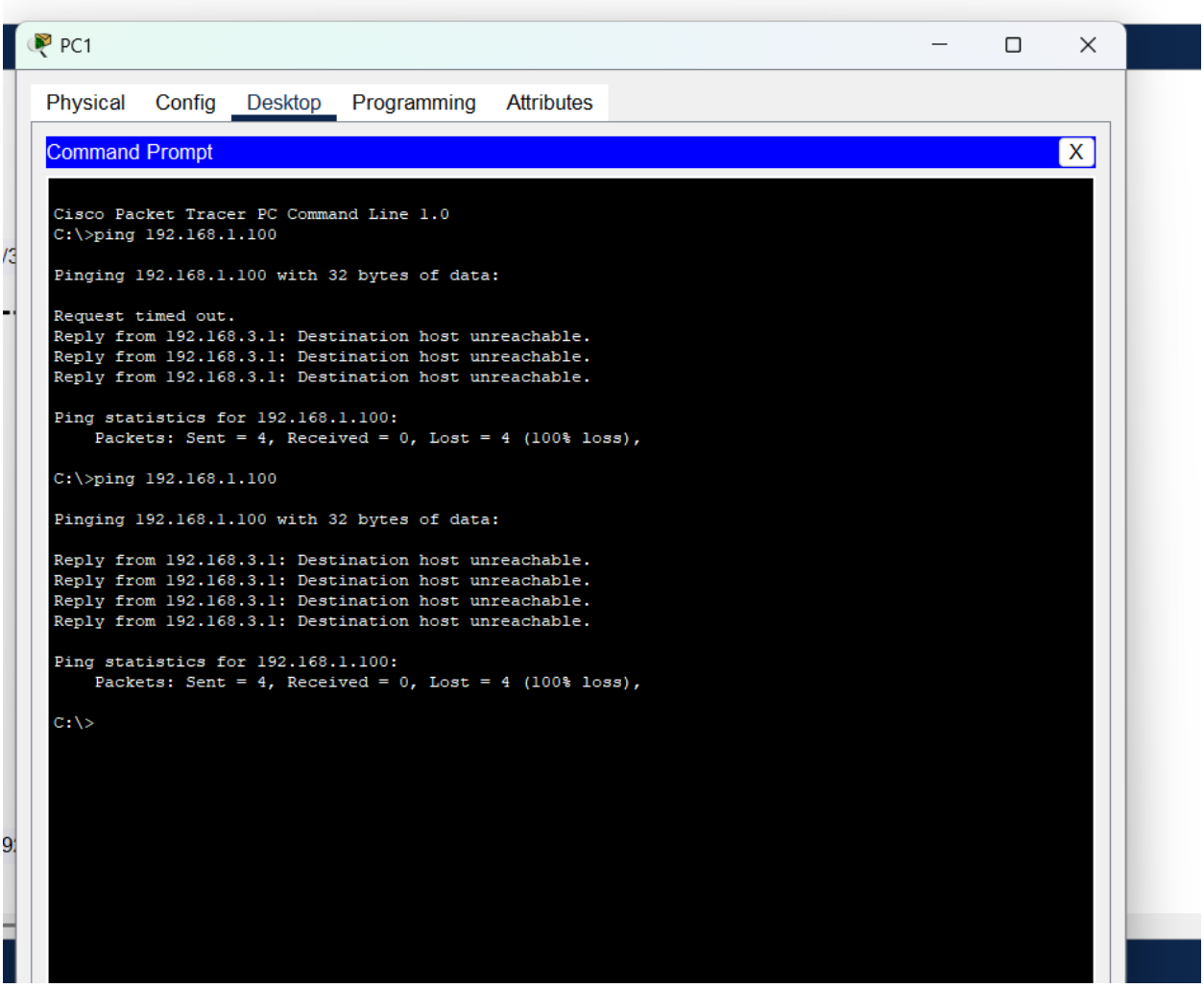
Pinging 192.168.2.101 with 32 bytes of data:

Reply from 192.168.2.101: bytes=32 time<1ms TTL=128
Reply from 192.168.2.101: bytes=32 time<1ms TTL=128
Reply from 192.168.2.101: bytes=32 time<1ms TTL=128
Reply from 192.168.2.101: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Type



Results and Observations

- Devices belonging to the restricted network were unable to ping or reach the target destination.
- Devices from permitted networks established successful connections.
- The extended access control list (ACL) was configured to permit traffic only on designated ports.

Conclusion

This lab effectively demonstrated the setup and validation of ACLs. Standard ACLs restrict traffic based solely on source IP addresses, whereas extended ACLs provide more

granular control by filtering based on protocols, ports, and destination IPs. Implementing ACLs enhances network security by regulating the flow of traffic through the router.

REFERENCES

1. YouTube – Standard Access List (ACL) for the Cisco CCNA Part 1, 2, and 3.
2. NetworkHmer Blog – Access Control Lists (Standard & Extended).

3. Cisco Systems. (2024). *Access Control Lists (ACLs) Configuration Guide, Cisco IOS Release 15.* Cisco. https://www.cisco.com/c/en/us/support/docs/ip/access-lists/26448-ACLsamples.html

4. Odom, W. (2020). *CCNA 200-301 Official Cert Guide, Volume 2.* Cisco Press. *(Chapters on Security Fundamentals and ACL Configuration.)*

5. Todd Lammle. (2020). *CCNA Certification Study Guide, Volume 2 (8th ed.).* Wiley. *(Provides examples of standard and extended ACL setups.)*